Chapter 18 exercises • The Linux system

OS18-E01 • Fork and exec

After fork, the child reads little memory and immediately execs.

1. How does COW reduce cost?
2. Which task receives the new image?
3. How must failed exec be handled?

OS18-E02 • Module references

A device module has one active file handle and one queued request.

1. Can it unload now?
2. Which events must finish first?
3. Can signatures prevent use-after-free?

OS18-E03 • VFS and page cache

Read returns 4 KiB from cache, then server power fails.

1. What visibility does read success prove?
2. Which states may remain undurable?
3. What does fsync change?

OS18-E04 • Credentials and regular-file DAC

V3.2 regular file, owner1000, group100, mode 0400, rw mount, no ACL/LSM denial. Operations are independent;capabilities apply in the corresponding user namespace.

1. Compare read/write/execute for fsuid1000 without capabilities, fsuid0 with DAC_OVERRIDE, fsuid0 without DAC capabilities, and fsuid2000 with DAC_READ_SEARCH only.
2. Why does mode 0401 change capable execute checking?
3. Does seteuid(1000) permanently drop authority from real1000/effective0/saved0?

OS18-E05 • Timeout, EIO, and late completion

R11, generation 3, is in flight with DMA toF40. One page 4096 B, no user copy yet, and one consumer reference remains.

Chapter 18 exercises • The Linux system (continued)

1. Can timeout immediately unmap/free? Give a counterexample.
2. After hardware stops and completes EIO, what are notification count and transferred bytes?
3. Handle duplicate same-generation completion and old3 against new4.
4. Can F40 be reused after terminal callback while consumer still holds it?

OS18-E06 • Clone sharing and ELF zero-fill

ParentTID100/TGID 100, fd4→O1. ELF PT_LOAD starts0x400000, filesz0x1800, memsz0x3000, page 4 KiB.

1. Count tasks, mm, fd tables, and original descriptions for fork versus VM/SIGHAND/THREAD/FILES clone.
2. How does v3.2 treat THREAD without SIGHAND, and SIGHAND without VM?
3. Calculate file/zero bytes per page. Does VMA creation imply residency?
4. Contrast exec failure before flush_old_exec and after its irreversible point.

OS18-E07 • IRQ reentry and unload races

CPU0 process holds L;its IRQ handler and CPU1 also use L. M retains one reference/handle or request, initially3+2.

1. Why can preemption disable alone still deadlock locally?
2. Does local IRQ disable exclude CPU1? Combine appropriate mechanisms.
3. Trace LIVE→GOING→close handles→finish requests→callback quiescence, with counts/free conditions.
4. Why is checking zero before withdrawing entry unsafe?

OS18-E08 • TCP segmentation and latency causes

IPv4/TCP, no options/offload, MTU1500, send accepts6000 B. Enqueue0, dispatch2, transmit complete3, ACK13 ms.

1. Give MSS, payloads, and L3 bytes excluding Ethernet/ACK/retransmissions.
2. Does acceptance prove remote processing?
3. Split queue, device, and subsequent ACK delay.
4. Another observation has queue20, retransmissions3, device errors1. Does it prove queue-caused tails? What more is needed?

Chapter 18 exercises • The Linux system (continued)

OS18-E09 • Pipe EOF, signals, and semop

After fork both processes retain writers. Child writesABC and closes its writer. A blocked standard signal occurs three times;semaphoresS0=1, S1=0.

1. Must EOF follow drainingABC? When does read return0?
2. Must the handler run three times? Contrast realtime signals.
3. In atomic semop(S0-=1, S1-=1), does S0 first become0 while S1 is insufficient?
4. Explain lost wakeup from checking before enqueue and the required correction.

References and attribution

Abraham Silberschatz, Peter B. Galvin, Greg Gagne. Operating System Concepts, 9th Edition.

Original educational synthesis by Lin Cing. Original material is licensed under CC BY 4.0. The referenced textbook is not distributed with these notes.

https://github.com/xinhi6716/computer-science-self-study

Lin Cing
https://github.com/xinhi6716/computer-science-self-study